

Particular and Homogeneous Solutions of $Ax = b$

A step-by-step walkthrough with two worked examples

1. The big idea

Most systems $Ax = b$ have more than one solution — often infinitely many. So instead of looking for *the* solution, you look for **all of them at once**. Here is the trick. Every solution can be split into two parts:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$$

$\mathbf{x}_p$ is any *one* solution you manage to find. It's called the **particular** solution because it's just one specific answer ($Ax_p = b$).

$\mathbf{x}_h$ is anything the matrix A sends to zero ($Ax_h = 0$). The set of all such vectors is called the **null space** of A .

Why this works: $A(\mathbf{x}_p + \mathbf{x}_h) = A\mathbf{x}_p + A\mathbf{x}_h = b + 0 = b$, so adding any null-space vector to $\mathbf{x}_p$ still gives a solution. And any two solutions differ by something the null space contains — so this formula covers every possible solution.

Picture it

Imagine the null space as a flat shape — a line, a plane, or a higher-dimensional version — passing through the origin. The set of solutions of $Ax = b$ is **that same flat, just slid over** so it passes through $\mathbf{x}_p$ instead of through 0. So $\mathbf{x}_p$ tells you where to start, and $\mathbf{x}_h$ tells you how to wander around the rest of the solution set.

2. The recipe (do this every time)

1 Row-reduce $[A \mid b]$

Glue A and b together into one big matrix $[A \mid b]$. Do row operations on it until you reach **reduced row-echelon form** (RREF). One row reduction — that's all you need.

2 Find the pivot columns and the free columns

Pivot columns have a leading 1 in the RREF. Their variables are **basic** — fully determined by the others. The other columns are **free** — you can pick any number for them. (Heads up: if the b -column itself becomes a pivot, the system has no solution. Stop there.)

3 Find $\mathbf{x}_p$: set every free variable to 0

With the free variables all set to zero, the basic variables are just whatever is on the right-hand side.

4**Find x_h : pretend the RHS is zero**

Now switch the free variables on one at a time. Make the first free variable equal to 1 (rest = 0) and read the basic variables. Then do it again with the next free variable. Each pass gives you one null-space basis vector.

5**Write down the general solution**

$x = x_p + c_1 v_1 + c_2 v_2 + \dots$ You get one free constant c_i for each free variable. Pick any numbers you like for them — you still get a solution.

3. First example · A 3×4 system

Here we have 3 equations and 4 unknowns. Let's walk through the recipe step by step:

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 7 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 5 \\ 9 \end{bmatrix}$$

Step A · Row-reduce $[A \mid b]$ **step 0**

Start: glue A and b together into the augmented matrix $[A \mid b]$

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 4 \\ 2 & 4 & 1 & 4 & 5 \\ 3 & 6 & 1 & 7 & 9 \end{bmatrix}$$

step 1

$$\begin{aligned} R_2 &\rightarrow R_2 - 2R_1 \\ R_3 &\rightarrow R_3 - 3R_1 \end{aligned}$$

Make the entries below the first pivot zero. Bonus: the same two operations also produce the column-3 pivot in row 2 for free.

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 4 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & -2 & -3 \end{bmatrix}$$

step 2

$$R_3 \rightarrow R_3 - R_2$$

Done — this is the RREF.

Blue = pivot columns, gold = free columns.

$$\begin{bmatrix} 1 & 2 & 0 & 3 & 4 \\ 0 & 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step B · Identify pivot and non-pivot columns

Pivot columns

columns **1** and **3** — basic variables x_1, x_3

Non-pivot columns

columns **2** and **4** — free variables x_2, x_4

Now turn each pivot row back into an equation:

$$\begin{aligned}x_1 + 2x_2 + 3x_4 &= 4 \\x_3 - 2x_4 &= -3\end{aligned}$$

Step C · Find x_p (set free variables to 0)

Make both free variables equal to 0. So put $x_2 = 0$ and $x_4 = 0$ into the equations above. They simplify to $x_1 = 4$ and $x_3 = -3$. That gives us our particular solution:

$$x_p = \begin{bmatrix} 4 \\ 0 \\ -3 \\ 0 \end{bmatrix} \quad \text{Check: } Ax_p = (4, 5, 9) = b \text{ [ok]}$$

Step D · Find x_h (solutions to $Ax = 0$)

Now pretend the right-hand side is zero (because we want vectors that A sends to zero). The equations become $x_1 = -2x_2 - 3x_4$ and $x_3 = 2x_4$. Now we switch each free variable on one at a time. First make $x_2 = 1$; then make $x_4 = 1$. That gives us two vectors:

Turn on x_2 : $x_2 = 1, x_4 = 0$

Then $x_1 = -2, x_3 = 0$

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

v_1

Check: $Av_1 = 0$ [ok]

Turn on x_4 : $x_2 = 0, x_4 = 1$

Then $x_1 = -3, x_3 = 2$

$$\begin{bmatrix} -3 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

v_2

Check: $Av_2 = 0$ [ok]

Step E · Assemble the general solution

$$x = \begin{bmatrix} 4 \\ 0 \\ -3 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

c_1 and c_2 are free — you can pick any numbers you want and the result is still a valid solution. Two free constants because there were two free variables.

4. Second example · A 4×5 system

Same recipe, slightly bigger system. This time 4 equations and 5 unknowns:

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 2 & 0 & 2 & 5 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 2 \\ 1 \\ 3 \\ 2 \end{bmatrix}$$

Step A · Row-reduce $[A \mid b]$

step 0

Start: the augmented matrix $[A \mid b]$

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 2 & 5 & 3 \\ 0 & 0 & 1 & 2 & 3 & 2 \end{bmatrix}$$

step 1

$$\begin{aligned} R_3 &\rightarrow R_3 - R_1 \\ R_4 &\rightarrow R_4 - R_2 \end{aligned}$$

Clear column 1 below R_1 and column 3 below R_2

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \end{bmatrix}$$

step 2

$$R_4 \rightarrow R_4 - R_3$$

Row 4 turned out to be the same as row 3, so it collapses to all zeros.

$$\begin{bmatrix} 1 & 2 & 0 & 1 & 3 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

step 3

$$\begin{aligned} R_1 &\rightarrow R_1 - R_3 \\ R_2 &\rightarrow R_2 - R_3 \end{aligned}$$

Clean up the entries above each pivot. This is the RREF.

Blue = pivot columns. Gold = free columns.

$$\begin{bmatrix} 1 & 2 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Step B · Identify pivot and non-pivot columns

Pivot columns	columns 1, 3, 4 — basic variables x_1, x_3, x_4
Non-pivot columns	columns 2, 5 — free variables x_2, x_5

Turn each pivot row into an equation:

$$\begin{aligned} x_1 + 2x_2 + x_5 &= 1 \\ x_3 - x_5 &= 0 \\ x_4 + 2x_5 &= 1 \end{aligned}$$

Step C · Find x_p (set free variables to 0)

Set both free variables to zero: $x_2 = 0$ and $x_5 = 0$. The equations now tell us $x_1 = 1$, $x_3 = 0$, $x_4 = 1$.

$$x_p = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{Check: } Ax_p = (2, 1, 3, 2) = b \text{ [ok]}$$

Step D · Find x_h (solutions to $Ax = 0$)

Pretend the right-hand side is all zeros. The equations become $x_1 = -2x_2 - x_5$, $x_3 = x_5$, $x_4 = -2x_5$. Switch one free variable on at a time — first x_2 , then x_5 :

Turn on x_2 : $x_2 = 1$, $x_5 = 0$

Then $x_1 = -2$, $x_3 = 0$, $x_4 = 0$

$$\begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

v_1

Check: $Av_1 = 0$ [ok]

Turn on x_5 : $x_2 = 0$, $x_5 = 1$

Then $x_1 = -1$, $x_3 = 1$, $x_4 = -2$

$$\begin{bmatrix} -1 \\ 0 \\ 1 \\ -2 \\ 1 \end{bmatrix}$$

v_2

Check: $Av_2 = 0$ [ok]

Step E · Assemble the general solution

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \\ -2 \\ 1 \end{bmatrix}$$

Same shape as before: a particular solution plus a free combination of null-space vectors. Again two free constants because there are two free variables.

5. Side-by-side comparison

	Example 1	Example 2
Size of A	3×4	4×5
Rank of A	2	3

Pivot columns	1, 3	1, 3, 4
Free variables	2 (x_2, x_4)	2 (x_2, x_5)
Dim of null space	$n - \text{rank} = 4 - 2 = 2$	$n - \text{rank} = 5 - 3 = 2$
$\mathbf{x}_p$	$(4, 0, -3, 0)$	$(1, 0, 0, 1, 0)$
Basis of Null(A)	$(-2, 1, 0, 0), (-3, 0, 2, 1)$	$(-2, 1, 0, 0, 0), (-1, 0, 1, -2, 1)$

Look at the pattern. In both examples, the number of free variables equals (number of columns of A) – (rank of A). The size of A by itself doesn't tell you this — only the rank does.

6. What to remember

- **Just one row reduction is enough.** You do it once on $[A \mid b]$, and both $\mathbf{x}_p$ and the null-space basis come out of the same RREF.
- **Pivot & free columns.** Pivot columns of A give you the basic variables. The other (non-pivot) columns give the free variables. The number of free variables = (number of columns) – (rank of A).
- **For $\mathbf{x}_p$:** set every free variable to 0, then read the basic variables from the RHS.
- **For $\mathbf{x}_h$:** pretend the RHS is zero, then turn each free variable on one at a time. Each turn gives you one null-space basis vector.
- **The general solution** is $\mathbf{x} = \mathbf{x}_p + c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots$ — one free constant for each free variable. You can pick any numbers for the c_i and still get a valid solution.
- **No-solution check.** If the RREF of $[A \mid b]$ contains a row of the form $[0 \ 0 \ \dots \ 0 \mid c]$ with $c \neq 0$, the system has *no* solution. (Equivalently: the b -column itself becomes a pivot column.)

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